In [2]:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from collections import Counter
import warnings
warnings.filterwarnings("ignore")
df = pd.read_csv("netflix_titles.csv")
print("Dataset loaded successfully!")
print("\nFirst 5 rows:")
display(df.head())
print("\nQ1: Dataset Shape")
print("Number of rows:", df.shape[0])
print("Number of columns:", df.shape[1])

# Q2: Display dataset information and data types

print("\nQ2: Dataset Information")
df.info()
print("\nData types:")
print(df.dtypes)

# Q3: Check missing values in each column

print("\nQ3: Missing Values")
print(df.isnull().sum())

plt.figure(figsize=(12, 6))
sns.heatmap(df.isnull(), cbar=False, cmap="viridis")
plt.title("Missing Values in Netflix Dataset")
plt.tight_layout()
plt.show()

# Q4: Analyze Movies vs TV Shows

print("\nQ4: Movies vs TV Shows")
type_counts = df["type"].value_counts()
print(type_counts)

type_counts.plot(
    kind="bar",
    figsize=(7, 5),
    color=["skyblue", "orange"])
plt.title("Number of Movies vs TV Shows")
plt.xlabel("Content Type")
plt.ylabel("Number of Titles")
plt.xticks(rotation=0)
plt.tight_layout()
plt.show()

# Q5: Analyze content added over the years

print("\nQ5: Content Added Over the Years")
df["date_added"] = pd.to_datetime(
    df["date_added"].astype("string").str.strip(),
    errors="coerce")

df["year_added"] = df["date_added"].dt.year
```

```
year_counts = df["year_added"].value_counts().sort_index()
print(year_counts)

year_counts.plot(kind="line", marker="o", figsize=(12, 5))
plt.title("Netflix Content Added by Year")
plt.xlabel("Year")
plt.ylabel("Number of Titles Added")
plt.grid(True)
plt.tight_layout()
plt.show()

# Q6: Analyze the distribution of release years

print("\nQ6: Distribution of Release Years")
print(df["release_year"].describe())
plt.figure(figsize=(12, 5))
sns.histplot(
    df["release_year"].dropna(),
    bins=30,
    kde=True)
plt.title("Distribution of Netflix Content by Release Year")
plt.xlabel("Release Year")
plt.ylabel("Number of Titles")
plt.tight_layout()
plt.show()

# Q7: Analyze Movie durations

print("\nQ7: Movie Duration Analysis")
movies = df[df["type"] == "Movie"].copy()

movies["duration_minutes"] = pd.to_numeric(
    movies["duration"].str.extract(r"(\d+)")[0],
    errors="coerce"
)

print("Movie duration statistics:")
print(movies["duration_minutes"].describe())

plt.figure(figsize=(10, 5))
sns.histplot(
    movies["duration_minutes"].dropna(),
    bins=30,
    kde=True
)
plt.title("Distribution of Movie Durations")
plt.xlabel("Duration (minutes)")
plt.ylabel("Number of Movies")
plt.tight_layout()
plt.show()

# Q8: Identify the most common genres

print("\nQ8: Most Common Genres")
genre_counts = (
    df["listed_in"]
    .dropna()
    .str.split(", ")
    .explode()
    .value_counts()
```

```
)

print("Top 10 genres:")
print(genre_counts.head(10))

genre_counts.head(10).sort_values().plot(
    kind="barh",
    figsize=(10, 6),
    color="teal"
)
plt.title("Top 10 Genres on Netflix")
plt.xlabel("Number of Titles")
plt.ylabel("Genre")
plt.tight_layout()
plt.show()

# Q9: Countries producing more Movies vs TV Shows

print("\nQ9: Countries by Movies vs TV Shows")
country_type_df = df[
    df["country"].notna()
    & (df["country"].str.strip() != "")
].copy()
country_type_df["country"] = country_type_df["country"].str.split(", ")
country_type_df = country_type_df.explode("country")
country_type_df = country_type_df.reset_index(drop=True)
country_type_df["country"] = country_type_df["country"].str.strip()
country_type_df = country_type_df[
    country_type_df["country"] != ""
].copy()
country_type_counts = pd.crosstab(
    country_type_df["country"],
    country_type_df["type"]
)

# Find the top 10 countries by total number of titles
top_countries = (
    country_type_counts
    .sum(axis=1)
    .sort_values(ascending=False)
    .head(10)
    .index)

top_country_types = country_type_counts.loc[top_countries]

print("\nTop 10 Countries: Movies vs TV Shows")
display(top_country_types)

top_country_types.plot(
    kind="bar",
    figsize=(12, 6)
)
plt.title("Movies vs TV Shows by Top 10 Countries")
plt.xlabel("Country")
plt.ylabel("Number of Titles")
plt.xticks(rotation=45, ha="right")
plt.legend(title="Content Type")
plt.tight_layout()
plt.show()
```

```python
# Q10: Analyze ratings by content type

print("\nQ10: Ratings by Content Type")

rating_type_counts = pd.crosstab(
    df["rating"].fillna("Unknown"),
    df["type"]
)

print(rating_type_counts)

rating_type_counts.plot(
    kind="bar",
    figsize=(13, 6)
)
plt.title("Content Ratings by Type")
plt.xlabel("Rating")
plt.ylabel("Number of Titles")
plt.xticks(rotation=45, ha="right")
plt.legend(title="Content Type")
plt.tight_layout()
plt.show()

# Q11: Find the oldest and newest titles

print("\nQ11: Oldest and Newest Titles")

valid_release_years = df.dropna(subset=["release_year"])

if not valid_release_years.empty:
    oldest_year = valid_release_years["release_year"].min()
    newest_year = valid_release_years["release_year"].max()

    print("\nOldest release year:", oldest_year)
    display(
        valid_release_years[
            valid_release_years["release_year"] == oldest_year
        ][["title", "type", "release_year"]].head(10)
    )

    print("\nNewest release year:", newest_year)
    display(
        valid_release_years[
            valid_release_years["release_year"] == newest_year
        ][["title", "type", "release_year"]].head(10)
    )
else:
    print("No valid release-year data available.")

# Q12: Analyze common genre combinations
print("\nQ12: Most Common Genre Combinations")
genre_combinations = (
    df["listed_in"]
    .dropna()
    .value_counts()
    .head(10)
)

print(genre_combinations)
```

```
genre_combinations.sort_values().plot(
    kind="barh",
    figsize=(12, 6),
    color="coral"
)
plt.title("Top 10 Genre Combinations on Netflix")
plt.xlabel("Number of Titles")
plt.ylabel("Genre Combination")
plt.tight_layout()
plt.show()
```

```
Dataset loaded successfully!

First 5 rows:
```

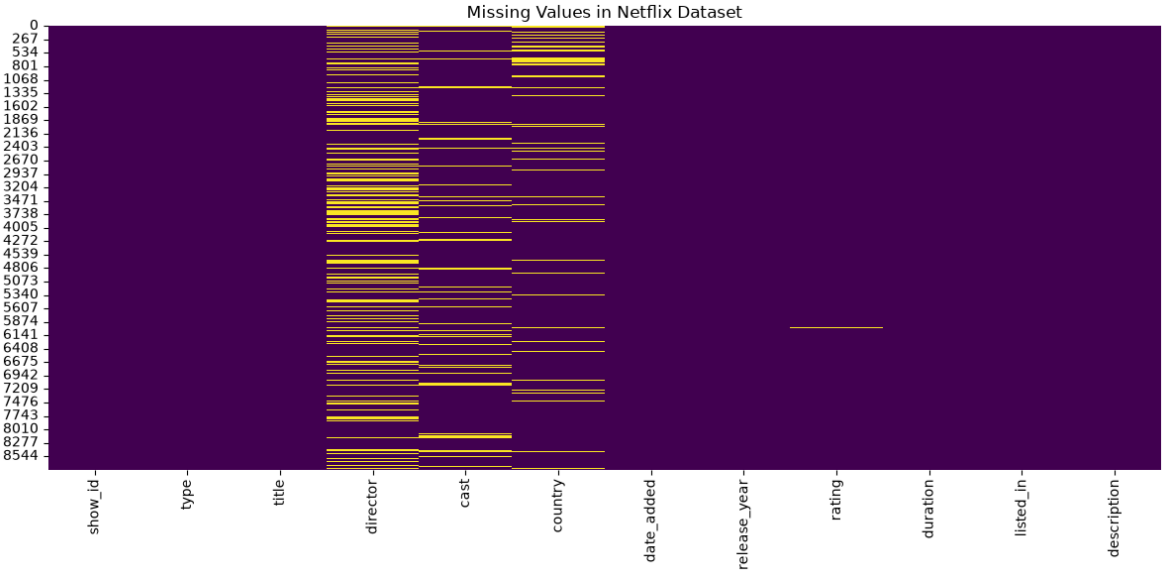
	show_id	type	title	director	cast	country	date_added	release_year	ra
0	s1	Movie	Dick Johnson Is Dead	Kirsten Johnson	NaN	United States	September 25, 2021	2020	P(
1	s2	TV Show	Blood & Water	NaN	Ama Qamata, Khosi Ngema, Gail Mabalane, Thaban...	South Africa	September 24, 2021	2021	
2	s3	TV Show	Ganglands	Julien Leclercq	Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi...	NaN	September 24, 2021	2021	
3	s4	TV Show	Jailbirds New Orleans	NaN	NaN	NaN	September 24, 2021	2021	
4	s5	TV Show	Kota Factory	NaN	Mayur More, Jitendra Kumar, Ranjan Raj, Alam K...	India	September 24, 2021	2021	

```
Q1: Dataset Shape
Number of rows: 8807
Number of columns: 12

Q2: Dataset Information
<class 'pandas.DataFrame'>
RangeIndex: 8807 entries, 0 to 8806
Data columns (total 12 columns):
 #   Column        Non-Null Count  Dtype
---  ------        --------------  -----
 0   show_id       8807 non-null   str
 1   type          8807 non-null   str
 2   title         8807 non-null   str
 3   director      6173 non-null   str
 4   cast          7982 non-null   str
 5   country       7976 non-null   str
 6   date_added    8797 non-null   str
 7   release_year  8807 non-null   int64
 8   rating        8803 non-null   str
 9   duration      8804 non-null   str
 10  listed_in     8807 non-null   str
 11  description   8807 non-null   str
dtypes: int64(1), str(11)
memory usage: 3.9 MB

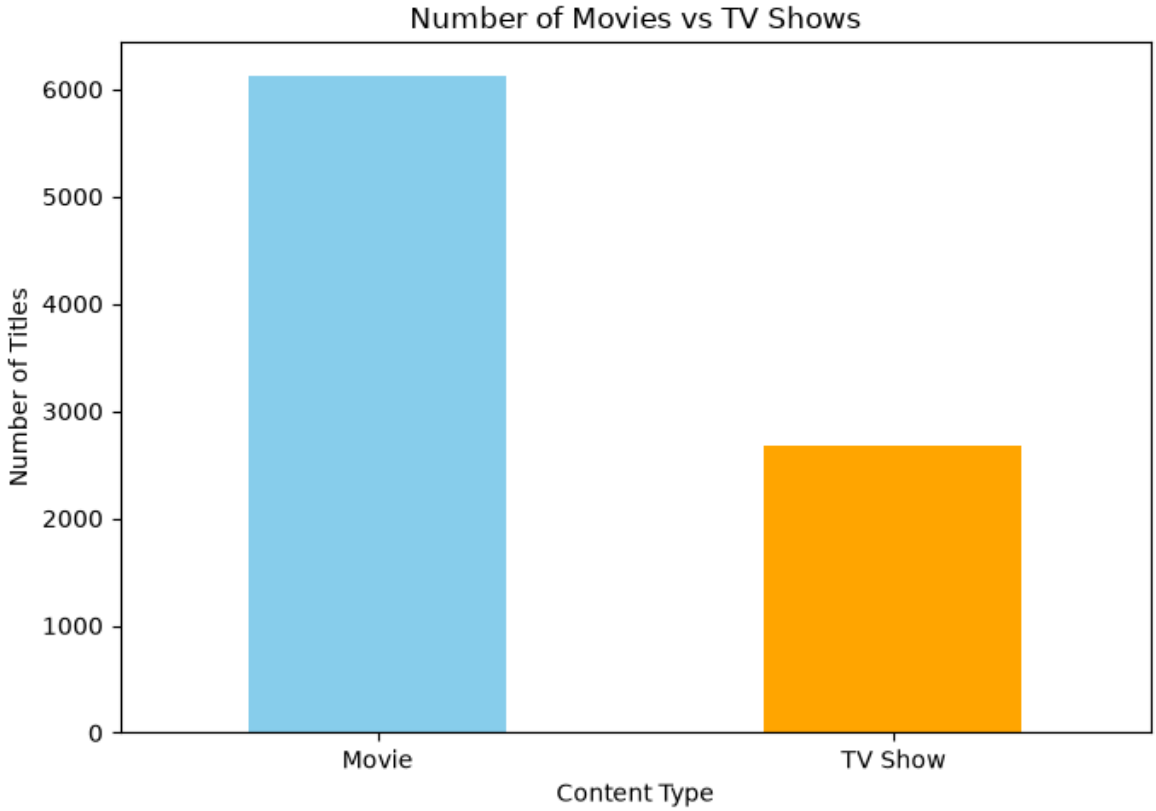
Data types:
show_id           str
type              str
title             str
director          str
cast              str
country           str
date_added        str
release_year    int64
rating            str
duration          str
listed_in         str
description       str
dtype: object

Q3: Missing Values
show_id            0
type               0
title              0
director        2634
cast             825
country          831
date_added        10
release_year       0
rating             4
duration           3
listed_in          0
description        0
dtype: int64
```



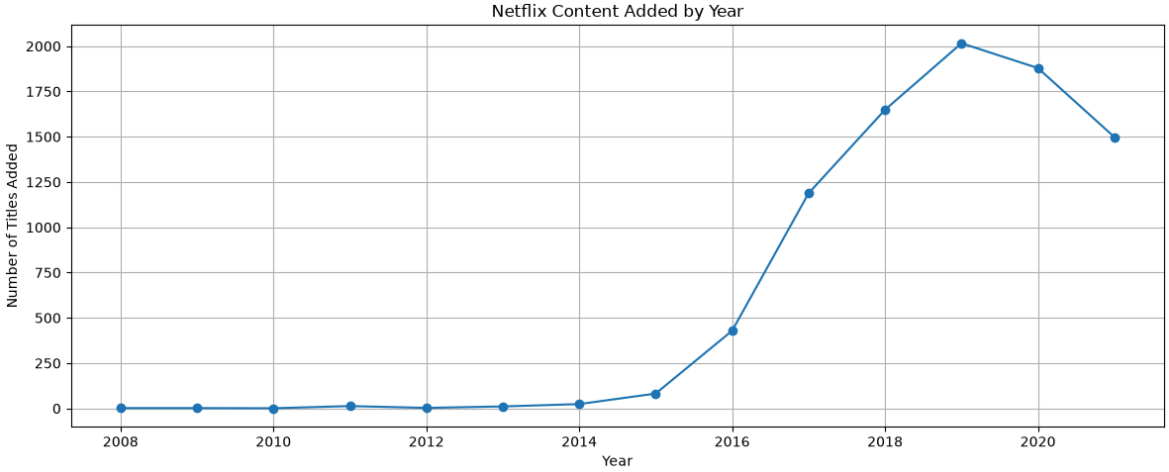
Q4: Movies vs TV Shows

```
type
Movie      6131
TV Show    2676
Name: count, dtype: int64
```



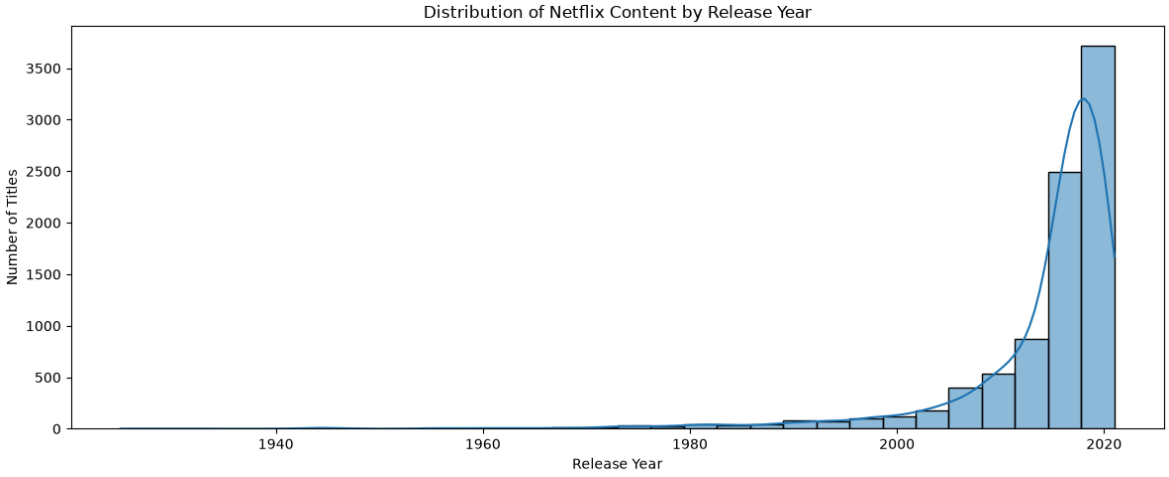
Q5: Content Added Over the Years

```
year_added
2008.0       2
2009.0       2
2010.0       1
2011.0      13
2012.0       3
2013.0      11
2014.0      24
2015.0      82
2016.0     429
2017.0    1188
2018.0    1649
2019.0    2016
2020.0    1879
2021.0    1498
Name: count, dtype: int64
```

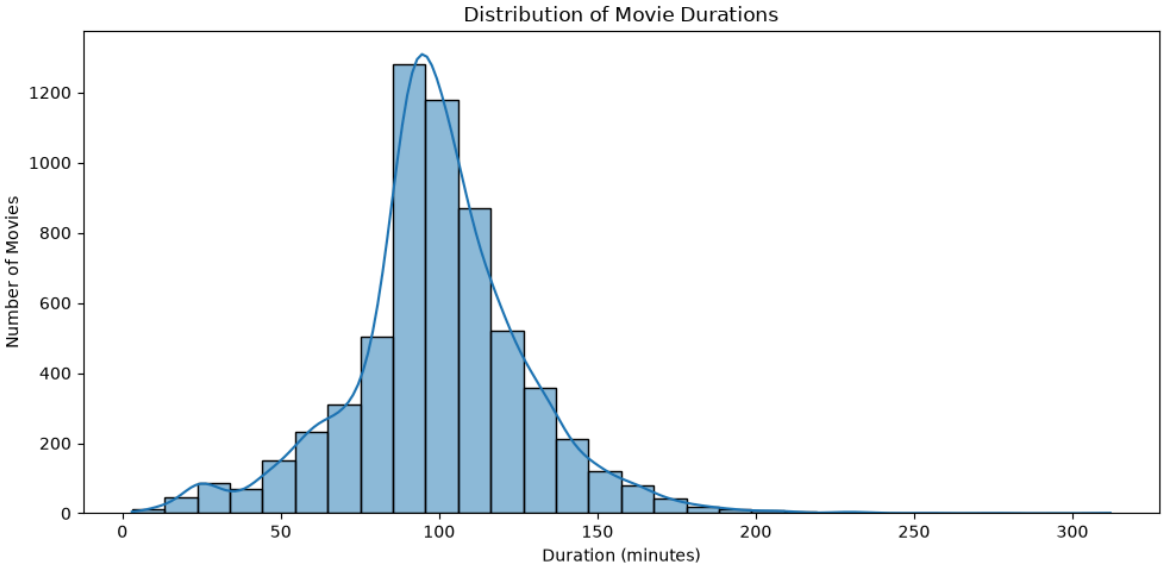


Q6: Distribution of Release Years

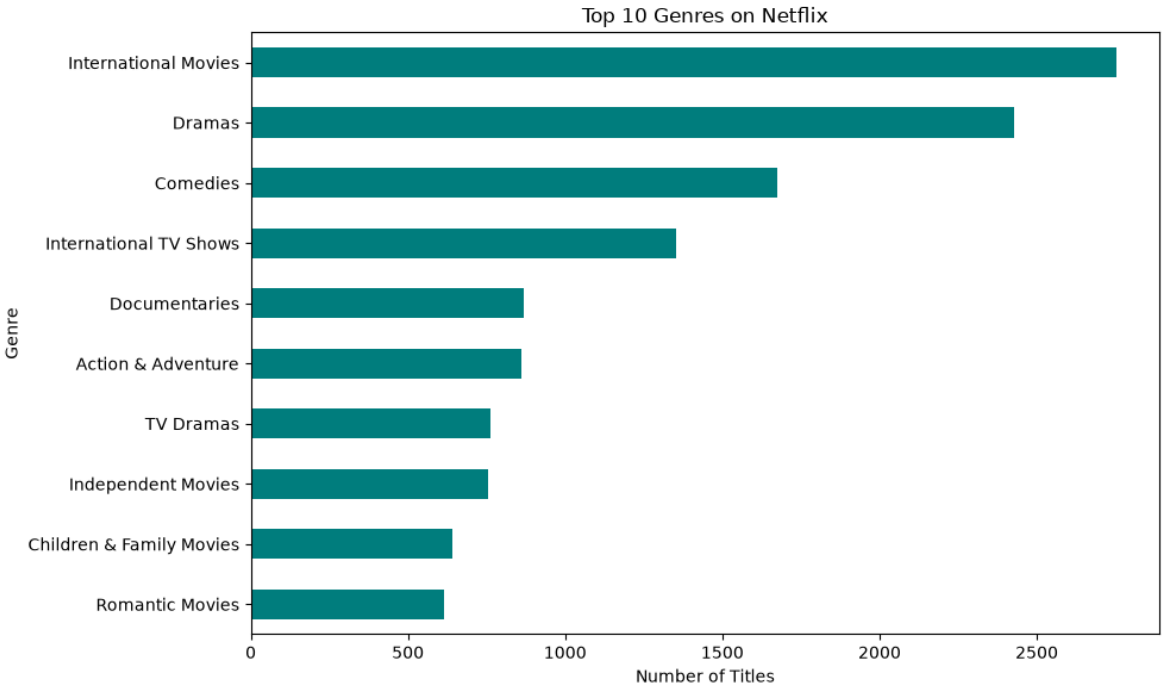
```
count    8807.000000
mean     2014.180198
std         8.819312
min      1925.000000
25%      2013.000000
50%      2017.000000
75%      2019.000000
max      2021.000000
Name: release_year, dtype: float64
```



```
Q7: Movie Duration Analysis
Movie duration statistics:
count    6128.000000
mean       99.577187
std        28.290593
min         3.000000
25%        87.000000
50%        98.000000
75%       114.000000
max       312.000000
Name: duration_minutes, dtype: float64
```



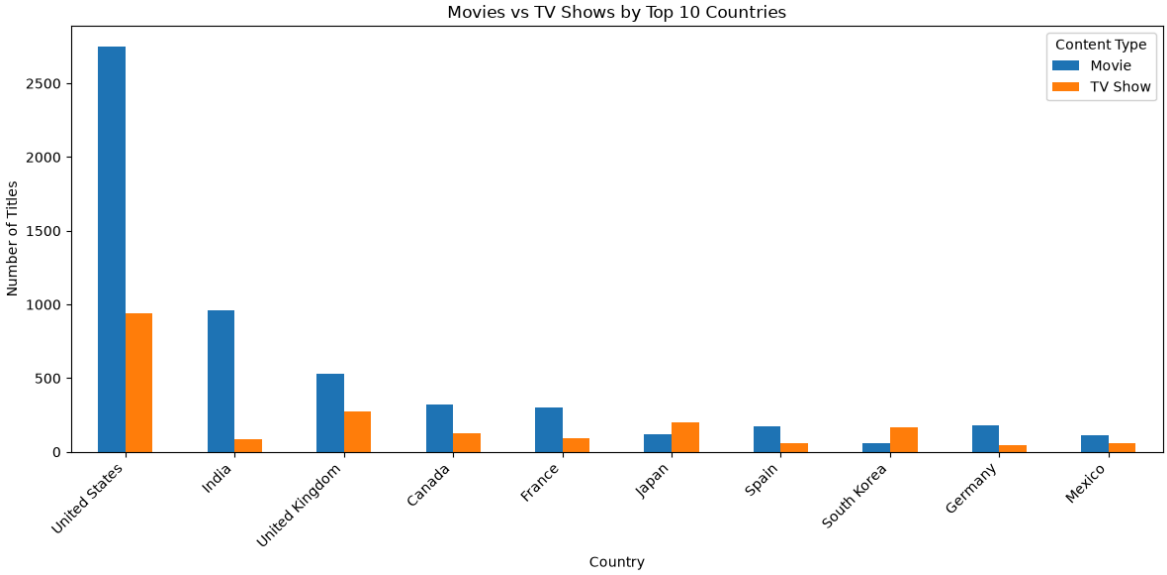

```
Q8: Most Common Genres
Top 10 genres:
listed_in
International Movies        2752
Dramas                      2427
Comedies                    1674
International TV Shows      1351
Documentaries                869
Action & Adventure           859
TV Dramas                    763
Independent Movies           756
Children & Family Movies     641
Romantic Movies              616
Name: count, dtype: int64
```



Q9: Countries by Movies vs TV Shows

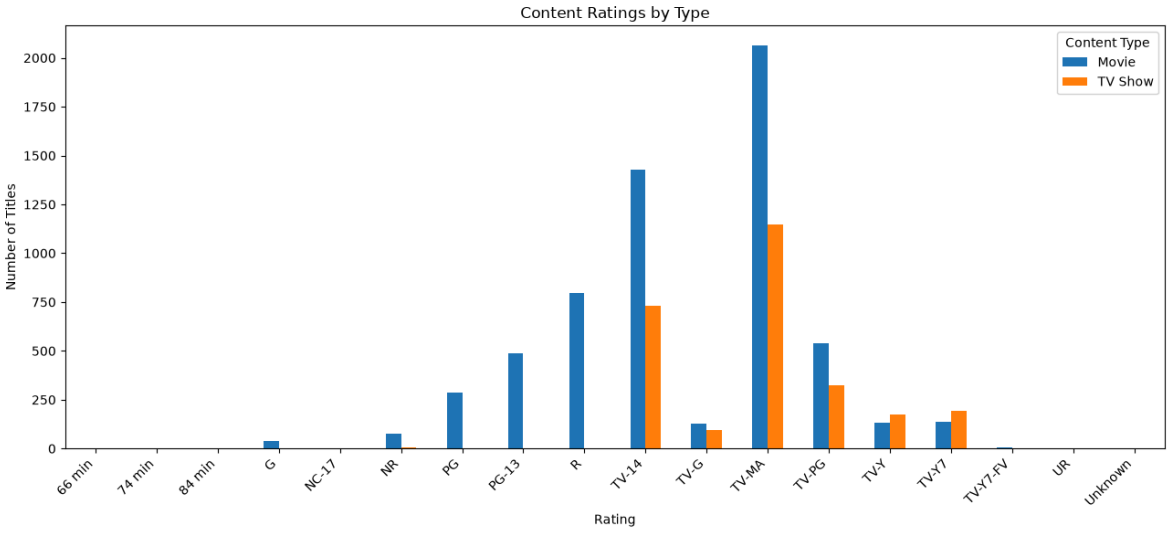
Top 10 Countries: Movies vs TV Shows

type	Movie	TV Show
country		
United States	2751	938
India	962	84
United Kingdom	532	272
Canada	319	126
France	303	90
Japan	119	199
Spain	171	61
South Korea	61	170
Germany	182	44
Mexico	111	58



Q10: Ratings by Content Type

type	Movie	TV Show
rating		
66 min	1	0
74 min	1	0
84 min	1	0
G	41	0
NC-17	3	0
NR	75	5
PG	287	0
PG-13	490	0
R	797	2
TV-14	1427	733
TV-G	126	94
TV-MA	2062	1145
TV-PG	540	323
TV-Y	131	176
TV-Y7	139	195
TV-Y7-FV	5	1
UR	3	0
Unknown	2	2



Q11: Oldest and Newest Titles

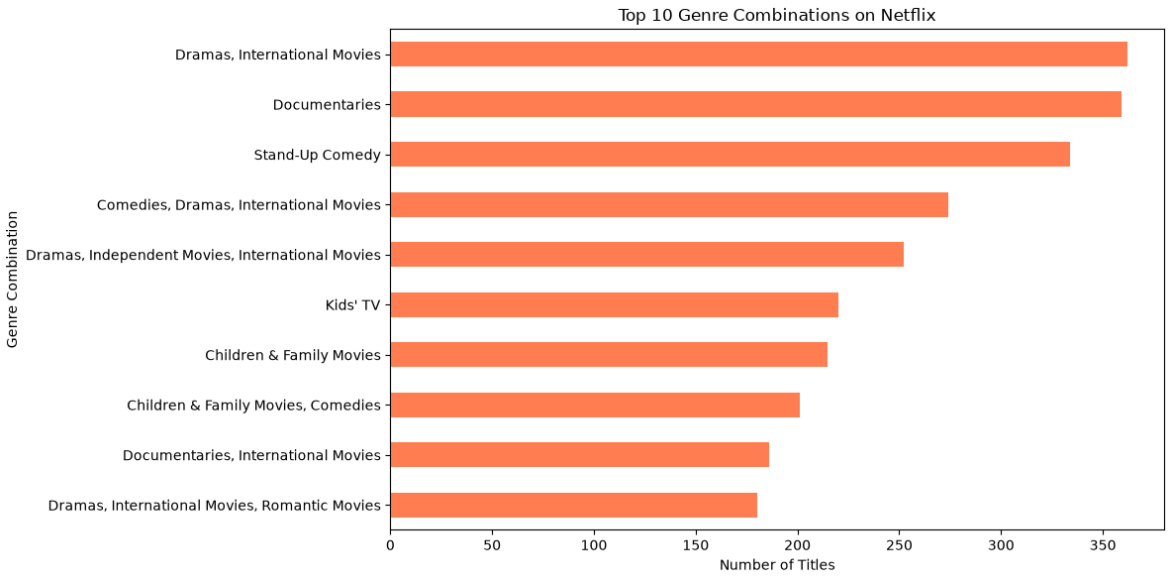
Oldest release year: 1925

	title	type	release_year
4250	Pioneers: First Women Filmmakers*	TV Show	1925

```
Newest release year: 2021
```

	title	type	release_year
1	Blood & Water	TV Show	2021
2	Ganglands	TV Show	2021
3	Jailbirds New Orleans	TV Show	2021
4	Kota Factory	TV Show	2021
5	Midnight Mass	TV Show	2021
6	My Little Pony: A New Generation	Movie	2021
8	The Great British Baking Show	TV Show	2021
9	The Starling	Movie	2021
10	Vendetta: Truth, Lies and The Mafia	TV Show	2021
11	Bangkok Breaking	TV Show	2021

```
Q12: Most Common Genre Combinations
listed_in
Dramas, International Movies                          362
Documentaries                                         359
Stand-Up Comedy                                       334
Comedies, Dramas, International Movies                274
Dramas, Independent Movies, International Movies      252
Kids' TV                                              220
Children & Family Movies                              215
Children & Family Movies, Comedies                    201
Documentaries, International Movies                   186
Dramas, International Movies, Romantic Movies         180
Name: count, dtype: int64
```



In [ ]: